JAYANTHA BS

SENIOR SOFTWARE ENGINEER - AI PLATFORMS, MULTI-AGENT RUNTIMES & BACKEND SYSTEMS

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PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience engineering high-performance AI agent runtimes, citation-aware RAG pipelines, and regulatory-compliant enterprise software. Proven track record architecting modular multi-agent frameworks using Python, FastAPI, and LangGraph, paired with high-volume vector search and real-time observability. Deep expertise in life-sciences compliance automation, including Veeva Vault MLR ingestion, Italian AIFA regulatory guardrails, and Salesforce auto-triggered CRM delivery. Adept at translating complex enterprise domain logic into resilient distributed architectures with strict state tracking, tool sandboxing, and sub-second retrieval latency.

CORE COMPETENCIES

AI & Agent Platforms: Multi-Agent Orchestration (LangGraph, LangChain), Context Management, Agentic Middleware, Tool Sandboxing
Regulatory & Compliance AI: AIFA Regulatory Governance, MLR Policy Guardrails, Audit Trail Grounding, Veeva Vault, Salesforce CRM
Search & Document AI: Vector Retrieval (Azure AI Search), Data Labs, Azure Vision, Citation Tracking, Layout Chunking, Metadata Enrichm
Backend & Cloud: Python 3.12, FastAPI, Django Ninja, PostgreSQL, Redis, REST & Streaming APIs, Docker, Kubernetes, Azure Architectur

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PROFESSIONAL EXPERIENCE

Senior Full Stack Developer | Phamax Consulting (Ariya Division) **Aug 2024 - Oct 2026 | Bengaluru, India**

AI AGENT PLATFORM & RUNTIME ARCHITECTURE

- Configurable Multi-Agent Runtime: Architected modular execution runtime using Python 3.12, FastAPI, LangGraph, and LangChain, integrating OpenAI and Anthropic models with agent-specific prompts, dynamic toolsets, and persistent storage.
- Zero-Downtime Dynamic Graph Compilation: Implemented hot-swappable graph compilation allowing agent capabilities to evolve.
- Streaming Execution Architecture: Developed thread-management and streaming-run APIs alongside AG-UI integration.
- Reusable Agent Middleware: Engineered model-call middleware for context summarization, token accounting, and loop-prevention.
- Enterprise Security & Tool Isolation: Implemented unified JWT authentication, SSO session propagation, and tool sandboxing.
- Distributed Observability & Evaluation: Integrated OpenTelemetry, OpenInference, and Arize Phoenix to trace agent lineages.

REGULATORY COMPLIANCE, INGESTION & DOCUMENT INTELLIGENCE

- AIFA Regulatory & Compliance-Aware Engine: Engineered deterministic compliance guardrails adhering to Italian standards (AIFA).
- Evidence Grounding & Citation Retrieval: Engineered citation-aware RAG pipeline utilizing Azure AI Search and bounding-box tracking.
- Ingestion Optimization: Reduced document-ingestion time by 50% by streamlining processing pipelines and HITL workflows.
- Enterprise Document Processing: Processed complex clinical decks/PDFs into Markdown/JSON via DataLabs and Azure Vision.

AI APPLICATIONS, ANALYTICS & ENTERPRISE INTEGRATIONS

- Veeva Vault & Salesforce Content Automation: Integrated Veeva Vault APIs within Content Wizard; automated Salesforce delivery.
- Content Wizard Platform: Built core platform using React, TypeScript, Django Ninja, and GrapesJS for rapid asset generation.
- Pharmaceutical Intelligence: Delivered two AI analytics suites empowering European pharma clients with market intelligence.
- Analytical Execution: Developed agent-driven analytical workflows integrating Microsoft Flint, Vega, and DuckDB.

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Software Developer | CamcomAI **Dec 2021 - Oct 2023 | Bengaluru, India**

AICAMERA & COMPUTER VISION PLATFORM

- AICAMERA Platform Architecture: Engineered core features of AICAMERA automated vehicle inspection platform, driving up to a 95% reduction in manual inspection turnaround and claim processing costs.
- Real-Time Orientation Classification: Implemented in-house vehicle-side classification model to guide users in angle capture.
- Edge Media & Telemetry Capture: Developed browser-based camera streaming via JavaScript MediaDevices APIs at 15 FPS video.
- CVAS Microservices Migration: Decomposed monolithic services into microservices and developed image-annotation tools (OpenCV).

Contract Developer | Medilenz and Risk Advisors Inc. **Oct 2020 - Nov 2021 | Bengaluru, India**

- Developed automated data-extraction pipelines using Python, Beautiful Soup, and Flask for structured medical claims information.
- Built interactive analytical dashboards and monitoring interfaces using ReactJS and Redux.

Software Engineer | EFI **Apr 2019 - Jul 2020 | Bengaluru, India**

- Automated enterprise directory management and distribution list synchronization via Python Active Directory/LDAP scripts.
- Developed monitoring capabilities for business-critical internal portals and built frontend components for Active Directory.

TECHNICAL SKILLS DIRECTORY

Programming Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5/CSS3
AI & Agentic Systems: LangGraph, LangChain, OpenAI API, Anthropic Models, RAG, Agent Middleware, Tool Orchestration, Context
Regulatory & MarTech: AIFA Compliance Governance, MLR Audit Trails, Veeva Vault (PromoMats API), Salesforce CRM / Marketing Cloud
Document Intelligence & Search: Azure AI Search, Azure Vision, DataLabs, Vector Embeddings, Chunking, Bounding-Box Attribution
Backend & Databases: FastAPI, Django, Django Ninja, Flask, SQLAlchemy, Alembic, PostgreSQL, MySQL, SQLite, Azure Cosmos DB, Ducl
Frontend & Visualization: React, TypeScript, Redux, Tailwind CSS, GrapesJS, FabricJS, Vega, Microsoft Flint
Cloud & DevOps: Microsoft Azure (App Services, Blob Storage), Docker, Kubernetes, CI/CD (GitHub Actions)
Observability & Dev Tools: OpenTelemetry, OpenInference, Arize Phoenix, GitHub Copilot, Cursor, Antigravity

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EDUCATION

Bachelor of Engineering (B.E.) - Medical Electronics (2015 - 2019)
M.S. Ramaiah Institute of Technology (MSRIT), Bengaluru, Karnataka, India